Dynamics of Autoimmunity - Small Project 1

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Abstract

This report details the computational modelling of an autoimmune response triggered by a viral infection. Using the COPASI software [1], we implemented a system of ordinary differential equations (ODEs) to describe the interaction between viruses, target host cells, and T lymphocytes [3]. Using both deterministic (LSODA [7], [8]) and stochastic (Direct Method [6]) solvers, we analysed the system's transition from a healthy state to one of chronic autoimmunity, which is characterised by recurrent relapses and remissions. The study highlights the fundamental differences between continuous and discrete modelling approaches in biology. This is particularly regarding the phenomenon of stochastic extinction of pathogens.

1 Introduction

Viruses have been linked to the onset, development, and worsening of several human autoimmune diseases. However, evidence also suggests that, in some cases, viruses may protect against autoimmune disease. Several mechanisms have been proposed to explain these effects [4]. The leading hypothesis is molecular mimicry, whereby microbial antigens share structural or sequence similarities with host autoantigens, thereby breaking immune tolerance.

Molecular mimicry refers to sequence similarity between foreign and self peptides. This has been recognised as one of the main mechanisms contributing to the breakdown of self-tolerance and the development of autoimmunity [2]. Since T-cell activation is a crucial step in mounting an immune response, any cross-reactivity at this stage can have severe pathological consequences [5].

In this project, we implemented the mathematical model proposed by Blyuss and Nicholson (2012) [3]. The authors modeled the dynamics of the immune response using the concept of tunable activation thresholds (TAT). According to this theory, a population of naive T cells (T_{in}) is activated into effector T cells (T_1) to combat viral infection (V) and infected cells (F). However, a fraction of these cells differentiates into autoreactive T cells (T_2) with a lowered activation threshold. These T_2 cells inappropriately target uninfected host cells (A_1, A_2), thereby initiating autoimmune disease. The model successfully captures normal viral clearance, chronic infection, and recurrent infections (relapses and remissions), depending on the viral infectivity rate.

2 Model

The model was implemented in COPASI [1] using a dimensionless formulation [3] based on fractions of the maximum carrying capacity. To enable successful stochastic simulations without encountering fractional particle limits or negative probabilities, a synthetic compartment volume of $V_{comp} = 10^5$ was defined, and the logistic growth terms were expanded into separate mass-action events. We used an interval size of 0.1 and set the random seed to 42 in each simulation.

2.1 Initial Values and Global Parameters

These initial values represent the immune system in its most basic state, immediately after a viral load has been introduced. The state variables and their initial concentrations (expressed as dimensionless fractions) are defined in the Table 1.

Variable	Biological Description	Initial Value
A_1	Primary target host cells (type 1)	0.9
A_2	Target host cells (type 2)	0.033
F	Infected host cells	0
T_{in}	Naive T cells	0.9
T_1	Effector (ability to kill F) T cells	0
T_2	Autoreactive (reduced activation threshold) T cells	0
V	Free viral particles	0.05

Table 1: Initial values of the system variables.

The constant transition rates, growth capacities and clearance rates that govern the system's dynamics are detailed in Table 2. The main bifurcation parameter is the viral **infectivity parameter**, λ, which is systematically varied during the computational experiments.

Parameter	Biological Description	Fixed Value
r_1, r_2	Logistic growth rates of target host cells A_1 and A_2	1.0, 0.5
p_1, p_2	Susceptibility factors of A_1 and A_2 to infection and autoimmunity	1.0, 1.0
N_p	Proportion of viral production attributed to target cell type 1	0.4
α_{act}	Activation rate of T_{in} into T_1 cells	100
α_F	Rate of destruction of infected cells by effector T cells	1.0
α_a	Rate of autoimmune destruction of host cells by T_2	50
μ_F	Natural death rate of infected cells	5.0
μ_1, μ_2	Death rates of effector (T_1) and autoreactive (T_2) T cells	0.4, 0.1
τ	Rate of threshold tuning (differentiation of T_1 into T_2)	0.005
k	Rate of viral production by infected cells	105
γ	Natural clearance rate of the virus	10
λ	Viral infectivity rate (bifurcation parameter)	Varied

Table 2: Global parameters used in the mathematical model.

2.2 List of Reactions

To prevent negative probabilities during stochastic simulations, terms of the form $r \cdot A \cdot (1-A)$ were mathematically expanded into linear production ($r \cdot A$) and quadratic density-dependent death ($r \cdot A^2$). This yields the following 20 reactions:

Name	Reaction	Rate Law
1. Production A_1	$\rightarrow A_1$; A_1	$r_1 \cdot A_1$
2. Overpopulation A_1	$A_1 \rightarrow$; A_1	$r_1 \cdot A_1^2$
3. Infection A_1	$A_1 \rightarrow$; V	$p_1 \cdot \lambda \cdot A_1 \cdot V$
4. Autoimmunity A_1	$A_1 \rightarrow$; T_2	$\alpha_a \cdot T_2 \cdot A_1$
5. Production A_2	$\rightarrow A_2$; A_2	$r_2 \cdot A_2$
6. Overpopulation A_2	$A_2 \rightarrow$; A_2	$r_2 \cdot A_2^2$

Name	Reaction	Rate Law
7. Infection A_2	$A_2 \rightarrow\ ;\ V$	$\lambda \cdot A_2 \cdot V$
8. Autoimmunity A_2	$A_2 \rightarrow\ ;\ T_2$	$p_2 \cdot \alpha_a \cdot T_2 \cdot A_2$
9. Creation F	$\rightarrow F\ ;\ A_1, A_2, V$	$\lambda \cdot (p_1 N_p A_1 + (1 - N_p) A_2) \cdot V$
10. Natural Death F	$F \rightarrow$	$\mu_F \cdot F$
11. T_1 kills F	$F \rightarrow\ ;\ T_1$	$\alpha_F \cdot T_1 \cdot F$
12. T_2 kills F	$F \rightarrow\ ;\ T_2$	$\alpha_a \cdot T_2 \cdot F$
13. Increase T_{in}	$\rightarrow T_{in}\ ;\ T_{in}$	T_{in}
14. Overpopulation T_{in}	$T_{in} \rightarrow\ ;\ T_{in}$	T_{in}^2
15. Activation $T_{in} \rightarrow T_1$	$T_{in} \rightarrow T_1\ ;\ F$	$\alpha_{act} \cdot T_{in} \cdot F$
16. Death T_1	$T_1 \rightarrow$	$\mu_1 \cdot T_1$
17. Change $T_1 \rightarrow T_2$	$T_1 \rightarrow T_2$	$\tau \cdot T_1$
18. Death T_2	$T_2 \rightarrow$	$\mu_2 \cdot T_2$
19. Production V	$\rightarrow V\ ;\ F$	$k \cdot F$
20. Clearance V	$V \rightarrow$	$\gamma \cdot V$

Table 3: A complete list of reactions, as implemented in COPASI.

3 Results

To properly visualize variables with vastly different concentration scales, the results for each simulation run are split into two separate plots: one displaying T_2 lymphocytes and infected cells (F) microscopic populations and another showing the fractions of T_1 lymphocytes. We conducted a parameter sweep over $\lambda \in \{0.1, 0.5, 1.0, 3.0\}$. For each scenario, we used a deterministic method - LSODA and a stochastic method -Direct Method based on the Gillespie algorithm on the right.

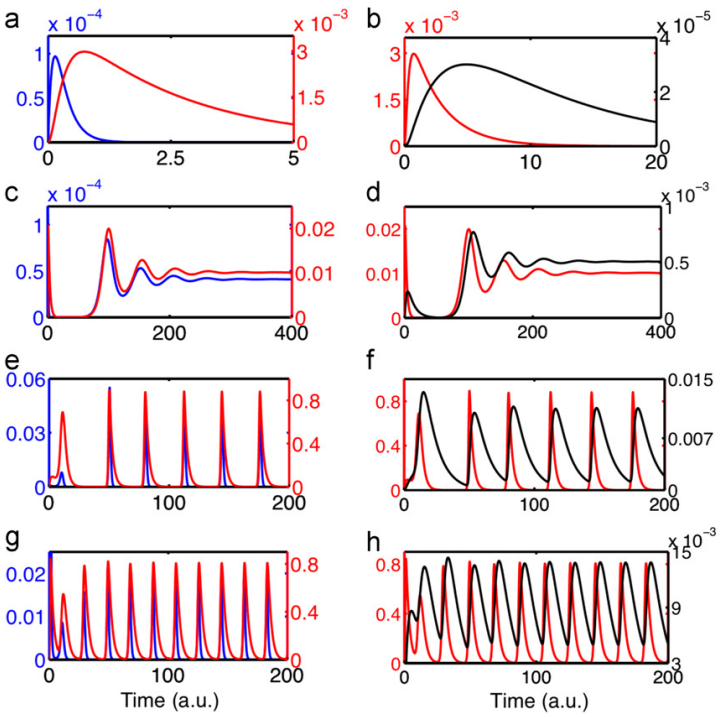

Figure 1: Reference numerical simulation results from the original publication. (a) and (b) $\lambda = 0.1$, (c) and (d) $\lambda = 0.5$, (e) and (f) $\lambda = 1.0$, (g) and (h) $\lambda = 3.0$. In all plots, the colours denote infected cells F (blue), activated T cells T_1 (red), and autoreactive T cells T_2 (black). Time is measured in arbitrary units.

3.1 Normal Viral Clearance ($\lambda = 0.1$)

At low infectivity, the viral load is rapidly suppressed. The stochastic model reveals definitive viral extinction. On the left are the graphs generated using LSODA, and on the right those generated using the stochastic method.

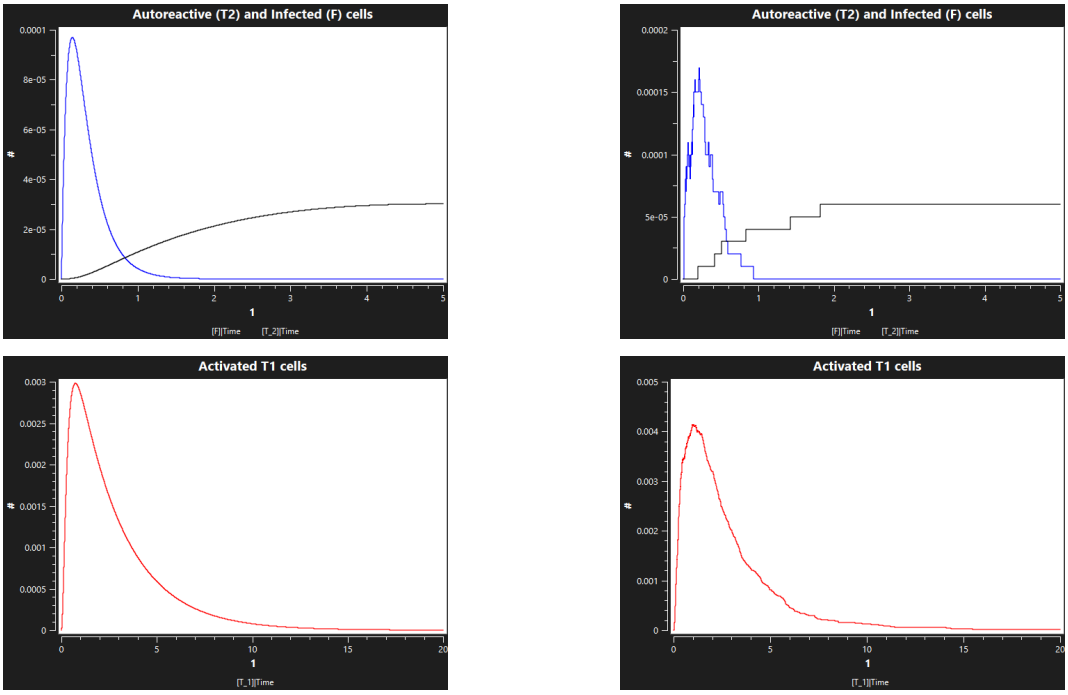

Figure 2: Infection dynamics for $\lambda = 0.1$. Rapid viral clearance.

3.2 Chronic Infection ($\lambda = 0.5$)

An intermediate viral infectivity leads to a stable chronic state with persistent target cell damage.

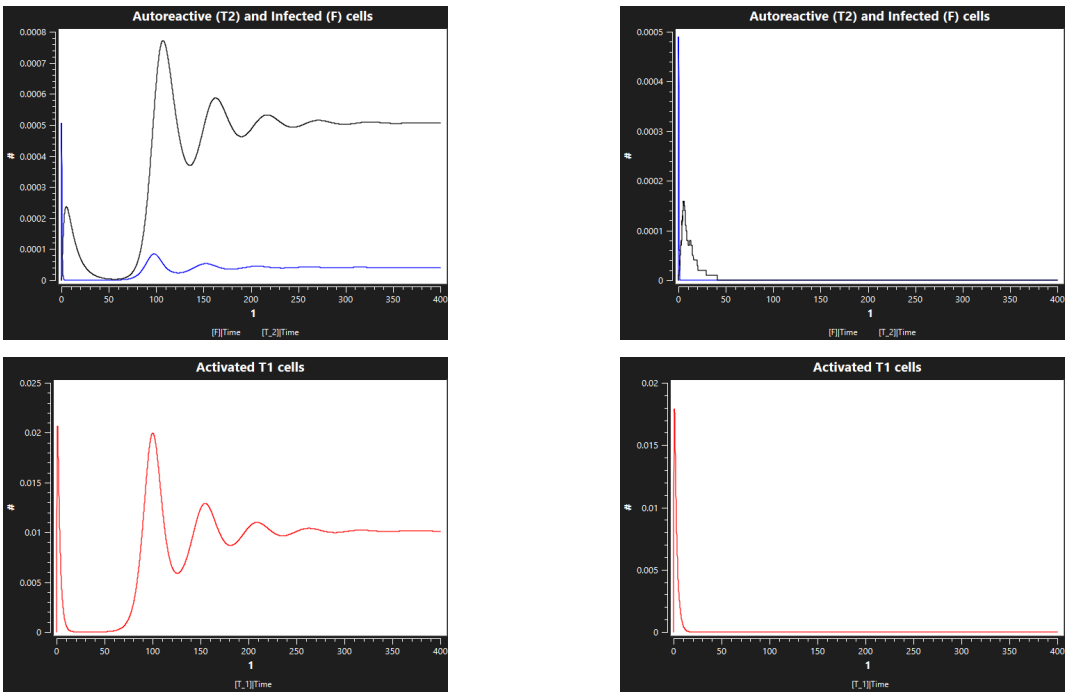

Figure 3: Infection dynamics for $\lambda = 0.5$. The system enters a stable chronic infection state.

3.3 Autoimmune Relapses and Remissions ($\lambda = 1.0$)

At $\lambda = 1.0$, the ODE system shows continuous relapses (limit cycle), while the stochastic system proves that the virus can be fully eradicated during remission, breaking the cycle.

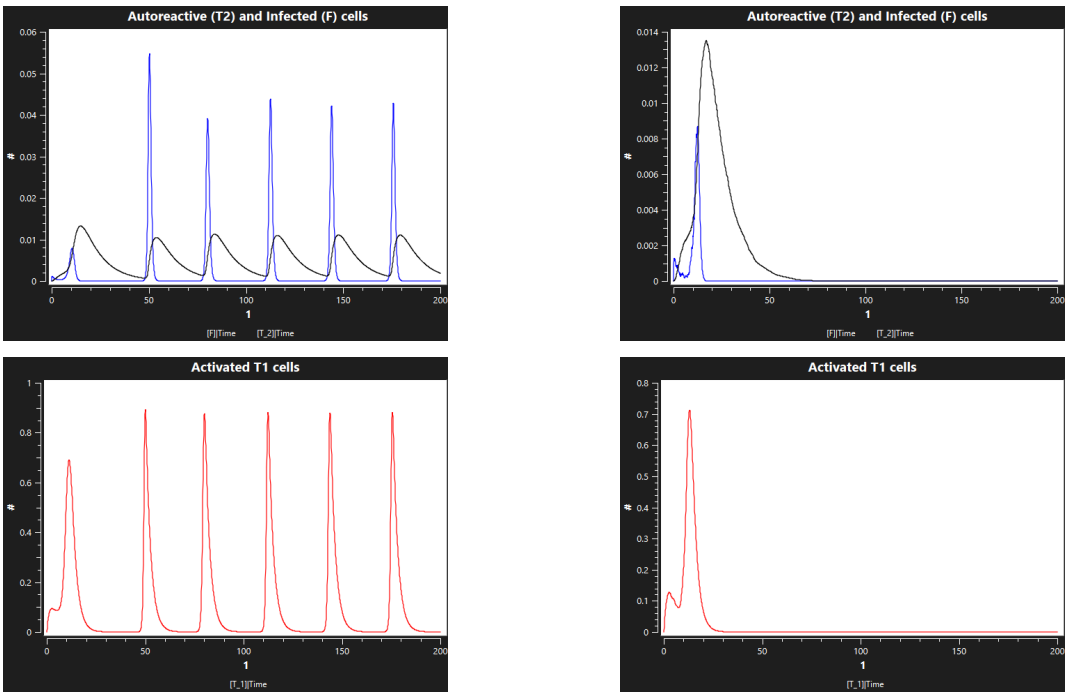

Figure 4: Infection dynamics for $\lambda = 1.0$. Deterministic oscillations vs. stochastic extinction.

3.4 High Infectivity ($\lambda = 3.0$)

Highly infectious viruses trigger an intense initial peak leading to damped oscillations and a severe chronic state in ODE.

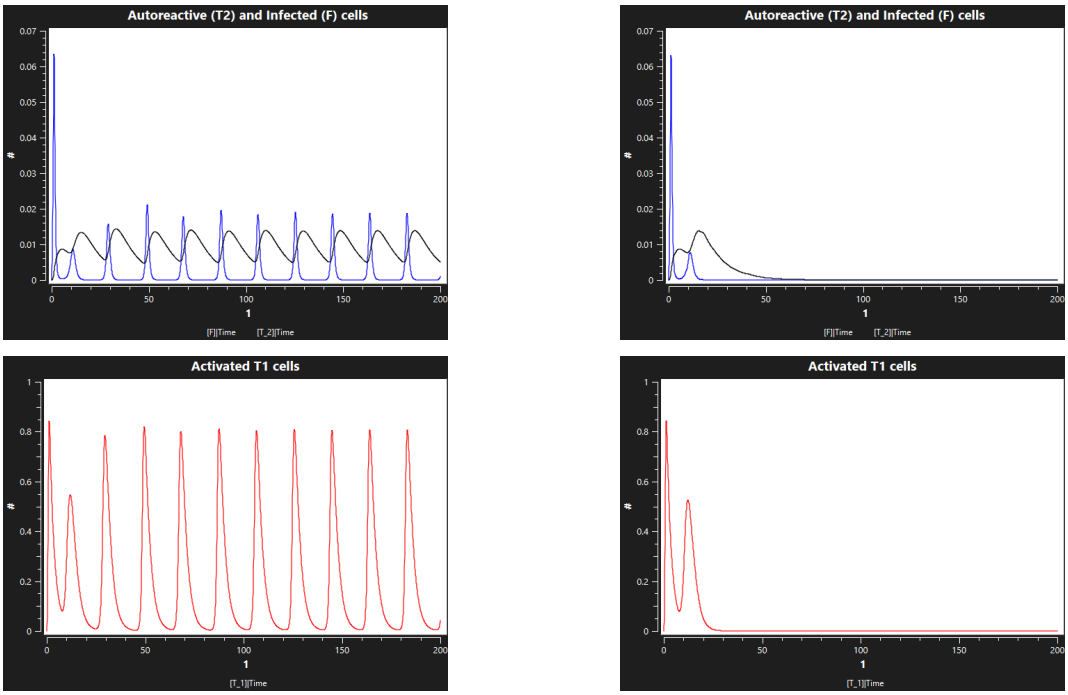

Figure 5: Infection dynamics for $\lambda = 3.0$. Damped oscillations and severe chronic state.

4 Discussion

Implementing the Blyuss and Nicholson model [3] successfully captures the intricate dynamics of virus-induced autoimmunity via molecular mimicry. By varying the viral infectivity parameter λ, we observed the full spectrum of disease progression, ranging from asymptomatic clearance to chronic damage and severe periodic relapses. Splitting the graphical results based on concentration scales enabled us to accurately observe the substantial expansion of T cells and the microscopic persistence of infected cells.

A key outcome of this project is the discrepancy between the deterministic and stochastic approaches, which is particularly evident in the oscillatory regime ($\lambda \in \{1.0, 3.0\}$). In the deterministic ODE framework, the variables are continuous. Consequently, the viral load never reaches absolute zero; it hides as an infinitely small fraction within the *valley of remission*, only to trigger another massive autoimmune relapse later.

Conversely, the stochastic model takes into account the discrete, particulate nature of biological entities. When the immune response is strong enough to suppress the virus, demographic noise often leads to the eradication of the last remaining viral particle (the absorbing state). Without the virus to continuously activate T_{in} cells, the production of F ceases, autoimmune T_2 cells die off, and the host fully recovers. Furthermore, charts generated using the stochastic method are less smooth and exhibit initial phase fluctuations. This demonstrates that, although deterministic models are ideal for analysing macroscopic trends and feedback loops, stochastic models are crucial for accurately predicting the elimination of pathogens and true biological recovery.

Author Contributions

Both authors contributed equally to completing this project. Jakub Giezgała focused primarily on the mathematical translation of the kinetic laws. He also managed the molecular mimicry theoretical background. In addition, he implemented the reaction network within COPASI. Piotr Kupidura was responsible for debugging the stochastic simulation parameters, performing the parameter sweep over λ and calibrating the output plots. He also wrote the final comparative analysis and revised the report. The `Autoimmunity.cps` file contains the parameters set for $\lambda = 3.0$.

References

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